

Subhranshu Sharma

Data Engineer | Machine Learning Engineer | Penetration Tester

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Professional Summary

Result-driven Machine Learning and Data Engineering enthusiast with hands-on experience building scalable **AI systems, ETL pipelines, and real-time data processing solutions** using Python. Skilled in developing end-to-end **deep learning models (TensorFlow, CNNs)** with strong foundations in **feature engineering, model training, and performance optimization**. Experienced in designing **structured data workflows, REST API integrations, and automation pipelines** for analytics-driven applications. Demonstrated expertise in **signal processing, computer vision, financial data analytics, and cross-platform deployment** of data-intensive systems. Passionate about transforming raw data into actionable insights through **analytics engineering, data visualization, and high-performance ML pipelines**, with a strong focus on scalable and production-oriented solutions.

Core Competencies

- Data Engineering & ETL Pipelines
- Machine Learning Model Development
- Deep Learning (CNNs, Neural Networks)
- Natural Language Processing (NLP Fundamentals)
- Real-Time Data Processing
- Signal Processing & Spectrogram Analysis
- API Integration & Automation
- Web Scraping & Data Extraction
- Model Optimization & Performance Tuning
- Data Visualization & Analytical Dashboards
- Cross-Platform Deployment (Linux, Android-Termux)
- Hardware Hacking & Embedded Systems

Technical Skills

- **Machine Learning & AI** : Supervised Learning, Deep Learning (CNN, RNN, LSTM), Transformer Models (Fundamentals), Feature Engineering, Model Training & Evaluation, Hyperparameter Tuning, Cross-Validation, Ensemble Methods (Random Forest, Gradient Boosting, XGBoost), NLP Preprocessing (Tokenization, TF-IDF, Word Embeddings), Signal & Spectrogram-Based Modeling, Data Cleaning & Transformation, ETL Pipelines, Real-Time Data Processing, Model Optimization & Performance Tuning, MLOps Fundamentals, Model Deployment Concepts
- **Languages**: Python (Advanced), Bash, C
- **Tools & Environments**: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, Librosa, Matplotlib, Git, Linux, Docker, APIs

WORK HISTORY

Remote Data Engineer (Contract) – Upwork

Lübeck, Germany | Oct 2022 – Nov 2022

- Worked under Harutyun Madoev (Bioinformatician at institute of neurogenetics, Luebeck) as data engineer.
- Used python (**Pandas**) to process csv file data according to requirements
- Automated data validation and cleaning pipelines to reduce manual processing time.
- Delivered structured, analysis-ready datasets aligned with research objectives.

Bug Bounty Hunting on HubSpot

February 2024 | [POC image](#) | [POC video](#)

- Used Burp Suite to find a vulnerability in HubSpot's chatbot API.
- Attacker could control the welcome messages sent from the other person's side while chatting with them.
- Attacker could overwrite chat history, making it seem like other person said something that they didn't.

PROJECTS

Eye-Gaze Controlled Cursor System (Computer Vision)

Dec 2021 | [working video](#) | [code](#)

- Developed real-time **Human-Computer Interaction (HCI)** system.

- Used TensorFlow-based facial landmark detection to detect eyes and eyebrows.
- Used **OpenCV Template Matching** for gaze-direction classification.
- Implemented real-time cursor movement using gaze-direction classification.
- Github repository starred by famous youtuber sentdex.

Tech Stack: Python, TensorFlow, OpenCV

Concepts: Computer Vision, Landmark Detection, Real-Time Inference

Audio-Based Keystroke Classification Using Deep Learning

November 2022 | [code](#)

- Built an end-to-end **audio signal processing + deep learning classification pipeline**.
- Implemented onset detection using **Librosa** to segment keystroke events.
- Converted 50ms audio windows into spectrograms for feature representation.
- Trained a **Convolutional Neural Network (CNN) using TensorFlow** on spectrogram.
- Achieved **68% classification accuracy vs 25% random baseline**.
- Demonstrated strong learning behavior with limited training data (39 seconds).

Tech Stack: Python, TensorFlow, Librosa, NumPy

Concepts: Signal Processing, Spectrogram Analysis, Deep Learning, Feature Extraction

Dumped RAM to Recover Password of a Hikvision Camera

Nov 2025 - Dec 2025 | [video explanation](#)

- To recover password of a Hikvision Camera.
- Extracted firmware from flash chip of embedded device for analysis.
- Reverse-engineered binary using Ghidra.
- Identified a pointer to plaintext password.
- Dumped RAM using gdbserver to recover the password.

Tech Stack: Ghidra, GDB, Assembly, Bash

Concepts: Hardware Hacking, Reverse Engineering, Firmware Modding, Memory Analysis, Embedded Systems

NSE Option Chain Scraper & Real-Time Visualization Engine

Nov 2024 – Feb 2025 | [screenshot](#) | [code](#)

- Developed a **real-time financial data scraping engine** for NSE Option Chain.
- Engineered automated data ingestion pipeline using Python.
- Designed interactive dashboard with:
 - Put & Call Open Interest (OI) plotter
 - Strike-price selection
 - Time-frame slider
 - Custom indicators
- Implemented caching layer to significantly reduce CPU load.
- Built cross-platform system, functional on Android (Termux).

Tech Stack: Python, Pandas, Matplotlib, Web Scraping

Concepts: Real-Time Data Engineering, Financial Data Analytics, Performance Optimization

Testimonials

Faraday Inc.: “Very responsible and competent, worked with a major vendor to resolve the issue he found”

ADDITIONAL TECHNICAL WORK

- Reported platform-level data persistence issue in Youtube.
- Built ultrasonic sensor-based obstacle avoidance car.
- Decoded PS/2 mouse signal using Arduino.
- Documented Docker-based workaround for building llama-cpp-python on Android.

Education

B.Tech. in Computer Science from VIT Bhopal University (2023 – 2027)

Certifications

- Google IT Support Professional Certificate (2026)
- National Cadet Corps – Certificate A